Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-II-Q03] | Part A (Q1-Q5)

A $10 \times 13 \times 14$ rectangular block loses ten slices, each one unit thick and parallel to one of its faces. Maximize the volume left after all slices.

尺寸 $10 \times 13 \times 14$ 的长方体，依次切去十片，每片厚一且平行于某个面。求剩余体积最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2011-II-Q09] | Part B (Q6-Q10)

Nonnegative $x_1, \ldots, x_6$ sum to one and satisfy $x_1x_3x_5 + x_2x_4x_6 \geq 1/540$. Let p/q be the greatest possible cyclic sum $\sum_{i=1}^{6} x_i x_{i+1} x_{i+2}$, with indices modulo six. Find $p + q$ in lowest terms.

非负数 $x_1, \ldots, x_6$ 的和为一，且 $x_1x_3x_5 + x_2x_4x_6 \geq 1/540$。设循环三项积和 $\sum_{i=1}^{6} x_i x_{i+1} x_{i+2}$ 的最大值为最简分数 p/q，下标模六，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-I-Q09] | Part B (Q6-Q10)

Triangle ABC has $AB = 40, AC = 31$ and $\sin A = 1/5$. It lies in rectangle $AQRS$ with B on side segment QR and C on side segment RS. Determine the maximum rectangle area. (A source-feasibility note accompanies the solution.)

三角形 ABC 满足 $AB = 40, AC = 31, \sin A = 1/5$，内接于矩形 $AQRS$，其中 B 在边线段 QR 上、C 在边线段 RS 上。求矩形的最大面积。（解析附来源可达性核查说明。）

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2022-II-Q06] | Part B (Q6-Q10)

Reals $x_1 \le \cdots \le x_{100}$ obey $\sum x_i = 0$ and $\sum |x_i| = 1$. If the maximum of $x_{76} - x_{16}$ is reduced m/n, find $m+n$.

实数 $x_1 \le \cdots \le x_{100}$ 满足 $\sum x_i = 0, \sum |x_i| = 1$。若 $x_{76} - x_{16}$ 的最大值为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2005-I-Q11] | Part C (Q11-Q15)

Find the largest possible diameter d of a semicircle contained in a side-8 square. If $d = m - \sqrt{n}$ with integers m, n, find $m + n$.

边长八的正方形内能容纳的半圆，其最大直径为 $d = m - \sqrt{n}$，其中 m, n 为整数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-I-Q14] | Part C (Q11-Q15)

Circle diameter AB has length eighteen. Extend AB past A to C, draw a tangent CT, and drop perpendicular AP to CT. If m is the greatest possible BP, find m^2.

圆的直径 AB 长十八，延长 AB 过 A 至 C，作切线 CT，从 A 向 CT 作垂足 P。若 BP 最大为 m，求 m^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2009-II-Q15] | Part C (Q11-Q15)

A circle has diameter $MN = 1$. On one semicircle, A is its arc midpoint and $MB = 3/5$. For C on the opposite semicircle, chords AC, BC meet MN at two points distance d apart. Its greatest value is $r - s\sqrt{t}$ in squarefree form. Find $r + s + t$.

圆的直径 $MN = 1$，一侧半圆弧中点为 A，该侧另有点 B 满足 $MB = 3/5$。C 在另一半圆弧上，AC, BC 与直径交点间距为 d，其最大值为 $r - s\sqrt{t}$，求 $r + s + t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-II-Q15] | Part C (Q11-Q15)

For $1 \le i \le 215$, set $a_i = 2^{-i}$, and set $a_{216} = 2^{-215}$. Positive reals x_i satisfy $\sum x_i = 1$ and $\sum_{i<j} x_i x_j = 107/215 + \sum_i a_i x_i^2/[2(1-a_i)]$. If the largest possible x_2 is reduced m/n, find $m+n$.

对 $1 \le i \le 215$，令 $a_i = 2^{-i}$，并令 $a_{216} = 2^{-215}$。正实数 x_i 满足 $\sum x_i = 1$ 及 $\sum_{i<j} x_i x_j = 107/215 + \sum_i a_i x_i^2/[2(1-a_i)]$。若 x_2 的最大值为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2024-I-Q15] | Part C (Q11-Q15)

Consider all rectangular boxes with surface area 54 and volume 23. Let r be the smallest radius such that every box can fit inside some sphere of radius r. Write $r^2 = p/q$ in lowest terms and find $p+q$.

考虑所有表面积为 54、体积为 23 的长方体。若半径为 r 的球能够分别容纳其中每个长方体，求最小的 r。将其平方写成最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-I-Q14] | Part C (Q11-Q15)

A convex pentagon $ABCDE$ has side lengths $14, 7, 24, 13, 26$ in order and interior angles $B = E = 60^\circ$. For arbitrary plane points X, the minimum of $AX + BX + CX + DX + EX$ is $m + n\sqrt{p}$ with squarefree p. Find $m + n + p$.

凸五边形 $ABCDE$ 的连续边长为 $14, 7, 24, 13, 26$，内角 $B = E = 60^\circ$。平面点 X 到五顶点距离之和的最小值为 $m + n\sqrt{p}$，p 无平方素因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-II-Q15] | Part C (Q11-Q15)

For exactly three positive parameters k, the function $f(x) = (x-18)(x-72)(x-98)(x-k)/x$ on $x > 0$ has its global minimum at exactly two distinct points. Find the sum of these k.

对恰好三个正参数 k，函数 $f(x) = (x-18)(x-72)(x-98)(x-k)/x$ 在 $x > 0$ 上的全局最小值恰在两个不同点取到。求这三个 k 的和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程